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VIETNAM NATIONAL UNIVERSITY, HO CHI MINH CITY

UNIVERSITY OF INFORMATION TECHNOLOGY

FACULTY OF INFORMATION SYSTEMS

FINAL PROJECT REPORT

STATISTICAL ANALYSIS

STATISTICAL AND MACHINE LEARNING ANALYSIS OF CUSTOMER RESPONSES TO PROMOTIONS AND DISCOUNTS

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Ho Chi Minh City, April, 2026

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ABSTRACT

The issue of how to attract customers no longer appears to be a primary issue for companies within the digital marketplace. Today's e-commerce companies face an additional challenge – that of optimizing their marketing strategies, so as to minimize the amount of waste incurred in their promotional efforts. This paper presents a solution to the above problem by presenting an analytical framework model, which consists of a combination of both inferential statistics and the latest machine learning techniques to help decode what is influencing customer loyalty and customer churn. This paper utilizes Lasso Regression analysis to identify the parameters that are actually affecting total spend, while using the DP-means cluster analysis technique to perform dynamic audience segmentation, in order to provide a clearer understanding of the way in which different audience segments respond to various promotions. Additionally, this report will detail a technical benchmarking study comparing two leading gradient boosting algorithms (LightGBM and CatBoost) in an effort to determine which is the superior predictive model for determining customer attrition. Based on the experimental findings, the hierarchy of influence between three types of behavioral signal (i.e., tenure, frequency of purchase, and history of recent transactions) has greater predictive capabilities than basic demographic information (i.e., gender and age). Another significant finding is that a "Heavy-Tailed" revenue distribution exists within the data (meaning there are a few individuals who create most of the profit for the business). This finding shifts the emphasis from a quantity-based marketing strategy to a value-based marketing strategy. Therefore, the results provided by this research serve as both a technical & strategic guide for e-commerce stakeholders who would like to shift their focus from using blanket-based discounting to utilizing highly tailored, behaviorally-based engagement.

1. Introduction

The fast development of the digital economy has significantly affected the retail industry and made e-commerce a very challenging platform for acquiring customers. Many companies have used various marketing techniques like flash promotions, personalized discounts, and cross-channel promotions to achieve revenue immediately. Nevertheless, considering the increasing customer acquisition cost (CAC), the current focus of business analysis has changed. According to Reichheld and Schefter (2000), even 5% of improvements in customer retention may lead to a profit increase of 25% to 95%. Hence, long-term customer relations matter more than a single purchase [1]. As a result, organizations are prioritizing CLV (Customer Lifetime Value) as a key performance metric. In this context, data analytics and churn prediction models have become essential tools to support sustainable business growth in the e-commerce sector [2].

The problem is that, while promotions and offers are extensively used, it is hard to determine whether they contribute to customer loyalty in the long run or just result in short-term sales along with further customer churn. Currently, businesses face three key challenges. First, to identify whether the customers continue to purchase even after the promotions or only respond to discounts. Second, while many organizations rely on generalized discounts, not every firm knows what particular customers or customer segments require targeted discounts. As a result, there is a misallocation of resources, and not all marketing investments can be considered effective. Third, even though companies gather extensive amounts of data about their customers, they fail to use data analytics to determine which clients will remain loyal to them in the long run or what kinds of promotions will maximize the company's gains. While there are many studies and business analyses focused on descriptive analytics and explaining performance metrics like overall sales or the success of certain promotion campaigns, fewer works explore more advanced analytics and predictions for future behavior through inferential methods and machine learning. This research fills a gap in the literature and makes an important contribution to improving business processes. The primary goal of this study is to analyze the efficiency of the usage of coupons as a means of increasing the loyalty of customers among e-commerce businesses.

The goals of the study can be formulated as follows:

- Measuring the efficiency of coupons: Understanding the link between coupons and such loyalty-related characteristics as frequency of purchases and cashback.

- Modeling the churn risk: Building a machine learning model that will reveal the correlation between extensive coupon usage and increased churn after the end of the promotion period.
- Analyzing the impact of satisfaction: Revealing the influence of satisfaction on the impact of coupons on customer loyalty.

The proposed study focuses on studying the impact of coupon-based promotions on loyalty measures, which include repurchasing and churn rate. Two datasets from Kaggle have been used in this study. These include "E-commerce Customer Churn Analysis" and "Marketing Insights for E-commerce". The limitations of this study include the scope of the independent variables. These variables include purchase frequency, cashback, tenureship, and customer satisfaction. The business context is confined to the B2C e-commerce industry, where the results will be used by online retailers whose main strategy of customer retention includes couponing. Furthermore, the temporal scope of the study is limited to the academic year 2025-2026.

1. Literature review

In this modern digital era, the concept of customer loyalty has transformed from a preference toward a specific brand to repeat buying behaviour. According to a study by Tran Anh et al. (2022), in the Vietnamese e-commerce market, promotional methods such as coupons or free shipping are indispensable for driving intent [3]. Instead, our synthesis of findings reveals a conspicuous paradox. Coupons provide immediate revenue; however, Garnefeld et al. (2012) cautioned against using coupon-based promotions too often, as this could lead to a "deal-seeking" mindset whereby customers tend to buy the product only when it is discounted and will switch to other brands at low switching costs after the incentive disappears [4]. By amalgamating these viewpoints, it immediately implies a criterion for our study - the meaning of measure success from using coupons needs to be evaluated not at the transaction approaching by using it right now but whether this pattern will set long-term habits.

In order to optimize the effectiveness of coupon strategies, customer segmentation is a critical first step, as businesses cannot treat all customers the same way. Businesses can apply segmentation in order to identify what groups require discount coupons to promote sales and what groups are sufficiently loyal to not require frequent financial incentives. One common approach to segmenting individuals is through the use of RFM Analysis (Recency, Frequency, and Monetary). Syarah (2025) advanced on this concept through the use of K-Means clustering algorithm to assist in the detection of the life cycles of clients [5]. From their research findings, it has been observed that Recency (how long ago was the last purchase) is the key determinant of such individuals. In our team's opinion, leveraging RFM scores should not stop at describing data, but must be used to provide dynamic features to advanced machine learning architectures to capture the shifting loyalty of coupon users.

The transition from descriptive analytics to predictive analytics has redefined how businesses manage customer churn.Scientific proof for the superiority of modern machine learning algorithms like Random Forest and XGBoost was provided by Agbemadon et al. (2023) [6]. It has been noted that the algorithms handle complex non-linear relations

and patterns in a large electronic commerce data set. Traditional mathematical ways fail to see these patterns due to the lack of depth that modern machine learning algorithms have. Feature Engineering is still hailed as the topmost factor regarding the accuracy of the prediction model. Feature Engineering obtains a new set of facts from the people's history of using the coupon codes. The working group adheres to the strategy of Agbemadon et al. (2023) in building specific data points. These specific data points follow the frequency and the regularity of how individuals who purchase goods utilize coupon codes, so that the prediction model becomes stronger.

Though there are many pieces of literature on the prediction of customer churn (Agbemadon et al., 2023) and the psychological impacts of promotions (Tran Anh et al., 2022), a large research gap is left in the integration of promotional behaviour and churn prediction. First, the studies with Tran Anh et al. (2022) accord more importance to quality of service perceptions, while Agbemadon et al. (2023) accord priority to algorithms performance. This work closes this gap by merging customer satisfaction dimensions with transactional data on coupons, thus providing a holistic picture of the underlying reasons why a customer leaves. Second, many researchers only verify whether or not a customer used a coupon (Yes or No). They do not estimate a degree of a customer being addicted to price cuts in the study. We go deeper by studying the intensity of use. We want to find the 'breaking point' when a customer will leave the brand due to the cessation of receiving coupons ("coupons stop coming in"). Finally, there is the gap with the forecast per segment. They do not differentiate the groups to forecast on a segment level. However, following the Syahra, et al. (2025) ideas, different groups behave differently. Rather than just forecast the likelihood that a customer will churn, our model considers the segment to which their behavior pattern belongs (e.g., 'Deal-Seeker' vs. 'Loyalist') and predicts the impact of the segment on the probability of churn. This lets the business figure out exactly who needs a discount and who stays because they genuinely like the brand.

1. Methodology

3.1. Research design overview

A quantitative research approach was employed and we used secondary transaction data. Instead of handling data as isolated figures, we structured this study to bridge the gap between descriptive analytics, which establishes a baseline of current consumer behavior, and predictive analytics, which anticipates future churn risks. The investigation was conducted through two logically connected phases:

- Exploratory phase: Implementing RFM (Recency, Frequency, Monetary) modeling and clustering algorithms to identify the latent customer segments. This stage tries to reveal the segments that respond to coupon triggers that are not obvious.

- Validation phase: Statistical hypothesis testing, together with machine learning models of LightGBM and CatBoost to measure the impact of coupon-based promotional activities on customer loyalty.

3.2. Data sources

The ground for this study is the combination of two big enough datasets, taken from the Kaggle platform, providing a 360 view on both transactional mechanisms as well as consumer psychology. Each plays a different but complementary role:

- Dataset 1: Marketing Insights for E-Commerce [6]

These include transactional audit logs (Online_Sales.csv), demographic profiles (Customers_Data.csv), and comprehensive information about the promotional campaigns. The transactional logs (Online_Sales.csv), demographic profiles (Customers_Data.csv), and information on promotional campaigns detail constitute the primary source for calculating RFM metrics and constructing variables like the Coupon Ratio.

- Dataset 2: Ecommerce Customer Churn Analysis and Prediction [7]

The data in this dataset is a collection of 5,630 customers with a focus on their records on churn behavior. That kind of dendrogram, needed to deal with more than 5,000 records gives the necessary complexity for training the models to predict the likelihood of losing customers.

3.3. Data pre-processing and feature engineering

We use the correct imputation methods to deal with missing values. We use the IQR approach to get rid of severe outliers. This manner, the T-test and regression models don't get messed up.

The Coupon_Ratio variable, which shows the ratio of coupon-using orders to total transactions, is an important part of this work. This percentage is a big part of figuring out which customers are buying products because they genuinely want them and which ones are largely buying things because they're on sale. Also, the Tenure variable was made the same in both datasets so that the customer lifecycle could be seen in the same way.

3.4. Data analysis techniques

3.4.1. Descriptive statistics and RFM analysis

Before moving to advanced analyses, our study first performed descriptive statistics to build a baseline for customer behavior. The main focus at this step is the RFM model application. Its components are deemed essential to convert low-level transactional logs into high-level behavior profiles.
- Recency (R): The number of days since the last purchase to measure how relevant it is.

- Frequency (F): Total transactions made to determine active customers.

- Monetary (M): Total amount of money spent to identify high-value customers. RFM metrics are the core input for the clustering and prediction stages.

3.4.2. Inferential statistics

This research used a battery of statistical tests to ensure that the patterns that are witnessed herein are not a result of chance.

- The Independent t-test was used in order to determine the Average Order Value (AOV) for the two groups of Customers, namely those shopping with coupons and those without. To do this, the total order sum (or AOV) was noted for every Transaction_ID.

- The One-way ANOVA test gives a comparison between the sales of a particular line of Products when given three different Discount (10%, 20%, or 30%) amounts on each line; this will allow for a determination of how much discount should be given on each line to maximize sales for the retailer and the supplier.

- Chi-square analysis looks for any nonlinear correlations between the two variables in Dataset 2: customer complaints (Complain) and churn behaviour (Churn).

3.4.3. Lasso regression

Lasso regression was implemented to establish a solid benchmark and identify the primary drivers of customer loyalty. The main advantage of this model is its ability to filter data. In particular, it can automatically eliminate statistical noise by shrinking the weight of low-impact coefficients to zero. The end result is a highly trimmed down model that only allowed the "Gold features" to stand out. This way, the business is metrics focussed on "Gold features" such as purchase frequency that drive profit. It does so without being distracted by duplicate data.

3.4.4. Customer segmentation (DP-Means Clustering)

To decode the diversity of the customer base, our study implemented the DP-Means clustering algorithm on Dataset 1 to segment overall numbers of customers into different behavioral groups.

- Input parameters: The model is trained on standardized features, such as the RFM metrics (Recency, Frequency, Monetary) along with the Coupon_Ratio variable, standardized through StandardScaler to balance the influence of all variables on the clustering process.

- Model optimization: Instead of choosing the number of clusters like with the traditional K Means algorithm, DP-Means optimizes the Penalty Parameter (λ), and when a customer acts differently from the existing cluster (exceeds the λ), a new cluster will be automatically created. The result is that the algorithm was able to pick 8 distinct clusters that reflect the data, more thoroughly.

3.4.5. Predictive modeling and evaluation

A major feature of the methodology includes construction of predictive models that will predict future customers' actions. A method for creating synthetic sample datasets (SMOTE or "synthetic minority over-sampling technique") to address class imbalance between both classes would help reduce bias due to class-size differences. The two machine learning algorithms (LightGBM and Catboost) were both trained concurrently. LightGBM was the model used primarily due to its speed and ability to handle large datasets. CatBoost is well suited for use with categorical variables and low risk of overfitting.

3.5. Ethical considerations and methodological limitations

The research used a secondary data that is anonymized, but the group is still aware of the ethical considerations. Before its use in the analysis, all personally identifiable information (PII) has been anonymized, and thus the study is about behavior, not people. To maintain analytical integrity, model output was evaluated by its Confusion Matrix and the ROC-AUC curve, which provides a solid foundation of objective and reliable findings before translating them into strategic business recommendations.

Considering limitations, secondary data allows us to clearly see what customers did, but not fully understand why they did it. Important aspects as emotions, personal motivations or perceptions of a brand, remain out of limits of the dataset. Another point worth mentioning is the fact that analysis is conducted within a relatively “closed system.” This means that external forces such as economic fluctuations or aggressive marketing campaigns from competitors are not directly captured, though they may have a great influence on customer churn, in practice.

1. Implementation and experimentation

4.1. Descriptive statistics and RFM analysis

The first step in putting this study into action is to focus on Data Synthesis. This is because the first transaction data are typically not related and have a lot of noise. Because of this, our group's main goal is to figure out how each client behaves. We went from seeing individual transactions to seeing the whole picture of customer lifecycles by combining separate parts of data from the Online_Sales, Customer_Data, and Discount_Coupon columns.

4.1.1. RFM modeling and feature engineering

We choose to use the RFM (Recency, Frequency, Monetary) analytical methodology to measure how customers act. This is the standard way to find out what customers value, and many studies have shown that it works. But the typical RFM model still has significant problems when it comes to predicting churn in an e-commerce setting, where coupons are a common way to promote products. RFM merely shows the results of behaviours, not the reasons why those behaviours happened. For example, two consumers can have the same RFM scores, but they may not be as loyal: one chooses our brand because they really like it, while the other chooses us because they can get coupons. So, we used Feature Engineering to build a special variable that better measures consumer engagement:

- Recency (R): Simply counting the days since the last transaction. The higher the R number, the more likely it is that people will leave.

- Frequency (F): The number of different invoices. This measure is very important for telling the difference between people who buy things often and those who only buy things when they have coupons.

- Monetary (M): This is the amount of money left over after taxes and shipping costs. It shows the real value that a consumer brought to the firm.

```
snapshot_date = sales_clean["Transaction_Date"].max() + pd.Timedelta(days=1)

rfm_df = (
    sales_clean
    .groupby("CustomerID")
    .agg(
        Recency=("Transaction_Date", lambda x: (snapshot_date - x.max()).days),
        Frequency=("Transaction_ID", "nunique"),
        Monetary=("Transaction_Value", "sum")
    )
    .reset_index()
)
```

Recency is kept in its original form as tree - based and clustering models can capture non - linear relationships without requiring monotonic alignment. Frequency is computed as the number of unique transactions to avoid duplication bias. Monetary is the total transaction value so we use the sum function.

Because RFM features are used for clustering and may exhibit high skewness, a log transformation is required. Therefore, in the next step, the skewness of each column will be examined to determine which variables should be log-transformed.

	Feature	Skewness
2	Monetary	6.747279
1	Frequency	5.689494
0	Recency	0.454355

Both Monetary and Frequency are strongly right-skewed (skewness > 5.5), so extreme values are present. To reduce variance and improve distribution balance we used log transformation for both features.

- Coupon_Ratio (CR): This is the most important part of our research and what sets it apart. It is the number of orders that use a promotion code divided by the total number of orders.

The CR is a way to find out how sensitive people are to price changes and how much they depend on promotions. A CR of almost 1.0 suggests that the customer and the business don't have a strong relationship and only buy when there are coupons.

4.1.2. Exploratory data analysis (EDA) and visualization

We use exploratory data analysis (EDA) and visualisation approaches to find out how coupon use and long-term loyalty are related in a complicated way:

- Analysis of promotional dependency: Revenue distribution charts and transaction frequency plots help us compare the Coupon_Ratio to how often people buy things. The goal of this strategy is to find out if buyers return to buy items because they have a strong connection to the brand or because they can get coupons.

- How coupons affect churn behaviours: We looked at the relationship between coupon sensitivity and the churn distribution. These illustrations let us test our hypothesis: that the group of customers who use coupons a lot is more likely to leave when they don't have any.

- How well do category-level coupons work? Bar charts that compare the average discount rates across different categories show how addicted people are to discounts for different types of products. This gives you a visual way to tell which groups are good for long-term committed customers and which ones are good for short-term "deal-seekers."

4.1.3. Experimental output: A holistic view of customer profile

The ultimate outcome of the EDA and feature engineering process will provide a set of consumer demographics, transaction details, and their sensitivity to coupons, thus enabling us to get a complete picture of each level/degree of client loyalty – based on three key elements:

- R, F and M (Recency, Frequency and Money Value) will indicate whether a given consumer's connection to the retailer is at its current level of strength.

- The Coupon_Ratio will help us determine if they are a long-term loyal customer or if they are simply shopping when they have a coupon.

- Using the Tenure measurement we will also be able to evaluate how consumer loyalty and coupon utilisation changed during their time as a customer.

4.2. Inferential statistics

After completing the customer profile description, the next phase continues with inferential statistics. The primary objective of this step is to use mathematical validation to determine whether our observations of the impacts of coupons on customer loyalty were statistically significant or just a coincidence.

4.2.1. Independent T-test: The impact of coupons on Average Order Value

In this study, the t-test is employed to compare AOV between two groups: customers who utilized coupons and those who did not. The t-test will be used to determine how coupon-based incentives affect a customer's power to purchase.

The Null Hypothesis () states that Average Order Values (AOV) of customers using coupons will be equal to the Average Order Values (AOV) of those customers that do not use coupons.
The Alternative Hypothesis() states that Average Order Values (AOV) of customers using coupons will not equal the Average Order Values (AOV) of customers that do not use coupons.

The T-test was done as shown below:

```
group_used = transaction_aov[transaction_aov['Coupon_Status'] == 'Used']['Line_Revenue']
group_not_used = transaction_aov[transaction_aov['Coupon_Status'] == 'Not Used']['Line_Revenue']
t_stat, t_p = ttest_ind(group_used, group_not_used, nan_policy='omit')
```

This code groups customers based on the use of the coupons in two classes. Then, it makes a comparison of their transaction values via a t-test. The essential measure in this case is the p-value, which leads to defining the statistical significance (value = 0.05).

4.2.2. ANOVA: Efficiency analysis across discount tiers

The One-Way ANOVA is a great tool for examining the efficiency of offering different amounts of discount (i.e. 10%, 20%, 30%) when compared against each tier after applying coupons. This will allow us to identify the saturation point where the addition of additional discounts ceases to be of benefit. By analyzing the mean differences between several groups, we aim to verify whether increased discount percentages continue to provide an equivalent economic return or if the revenue growth starts to slow down.

4.2.3. Chi-square test: Linking service friction to retention

For the categorical analysis in Dataset 2, the Chi-square test of independence will be performed on the sample data. The analysis aimed to determine whether unsatisfied experiences while making a purchase is correlated to customer behaviors toward attrition (leaving). This is particularly important as it distinguishes between "true" customers who will stay regardless of the use of coupons and those who are loyal only because they use coupons. The research is based on the premise that service failures will diminish the loyalty generated through coupon programs. Therefore, by using mathematical methods, it shows whether a customer's complaint, and the decision to leave, are independent of each other and/or dependent on one another. So, although coupons can get a person to buy immediately, the most significant reason that a customer will remain is due to the service quality provided to them. Thus, the findings will assist companies in creating a loyalty strategy that is sustainable, and not solely reliant on constant coupon offers.

In the case of Chi-square, the following code was used:

```
contingency_table = pd.crosstab(df_ecommerce['Complain'], df_ecommerce['Churn'])
chi2, p_chi, dof, expected = chi2_contingency(contingency_table)
```

The action taken in this code is to establish a contingency table between complaint and churn after which Chi-square statistic is calculated. The p-value indicates whether the two variables are related.

4.3. Lasso Regression

In this phase, this research starts to build a predictive baseline model to discover core factors that contribute to customers'values. We do not use traditional regression models like Linear Regression or Logistic Regression, but we decided to apply Lasso Regression (Least Absolute Shrinkage and Selection Operator) to leverage its advantage of Feature Selection.

4.3.1. Experimental setup and objectives

This experiment is designed to classify customers based on actual transaction behavior and we focus on separating strategic signals from data noise.

- Target variable: A binary variable represents "High-value" customers, defined as those whose transaction frequency (Frequency) exceeds the median. Customers with a frequency above the median were labeled as High-Value (1), while the rest were labeled 0.

- Independent variables: The calculated RFM-C metrics were Recency, Frequency, and Monetary, as well as Coupon_Ratio.

4.3.2. Implementation mechanism

The implementation mechanism of the experimental workflow follows three main steps, as shown in the original code:

- Features standardization: Since Lasso is a distance-sensitive mechanism, we needed to standardize the features to a common scale (mean = 0; std=1) using StandardScaler so that high magnitude variables (e.g. Monetary) do not put an undue advantage on the distance-based ratio variables (e.g. Coupon_Ratio) in the model.

- L1-regularization: The model was trained with the L1 regularization parameter (α=0.01). The penalty term associated with L1-regularization shrinkage will force all non-essential variables' coefficients toward zero.

- Shrinkage / Feature Selection: The model will shrink features that do not provide useful information to the target variable, as a result of the L1-penalty/regularization, which will prevent overfitting of Lasso due to giving less weight to insignificant predictors.

The model was trained as shown in the code below:

```
y = (rfm['Frequency'] > rfm['Frequency'].median()).astype(int)
X_train, X_test, y_train, y_test = train_test_split(X_scaled, y, test_size=0.2, random_state=42)

# Training Lasso for Feature Selection
lasso = Lasso(alpha=0.01)
lasso.fit(X_train, y_train)
```

In this code:

- Target variable y is developed in accordance to Frequency (high-value customers).
- Information is divided into training and testing data.
- The strength of regularization is controlled by a parameter alpha = 0.01.

To show the importance of features:

```
lasso_coefs = pd.DataFrame({
    'Feature': ['Recency', 'Frequency', 'Monetary', 'Coupon_Ratio'],
    'Coefficient': lasso.coef_
}).sort_values(by='Coefficient', ascending=False)
```

This demonstrates the important features to be used in prediction.

4.3.3. Statistical results and feature importance

The Lasso model achieved an impressive AUC-ROC score of 0.98, confirming that a simplified, sparse model can achieve nearly perfect classification accuracy.

Feature	Lasso Coefficient	Analytical interpretation
Frequency	+0.2227	The most critical factor. Repeat purchase behavior is the absolute strongest indicator of a High-Value Customer.
Recency	-0.0555	The longer the time since the last purchase, the lower the probability of being a High-Value customer.
Monetary	0.0000	Lasso determined this as redundant because its variance was already captured by Frequency, preventing multicollinearity.
Coupon_Ratio	0.0000	In this baseline, coupon dependency was treated as secondary noise compared to pure transaction frequency.

Table 4.1: Lasso Regression Coefficients Analysis

4.4. Segmenting customers using DP-Means Clustering

When parameters are verified through inferential statistics, the study comes to the phase of automatically dividing customers into groups. We aim to replace subjective confirmation with a segmentation model that uses real data. Consequently, customer profiles are identified based on purchasing behaviors and levels of coupon sensitivity.

4.4.1. Data preprocessing and feature scaling

The DP-Means algorithm requires the input metrics, namely: Recency (R), Frequency (F), Monetary (M), and Coupon_Ratio, to be Z-Score standardized before running. The reason for this is that each of these metrics is expressed in a different unit of measure (e.g., days for Recency, counts for Frequency, currency for Monetary and a percentage for Coupon_Ratio) and needs to be converted to a similar unit. This allows us to limit the effect of the variable with the highest numerical range (Monetary) on the calculation of distance metrics and, therefore, the clustering results.

4.4.2. Optimizing the number of clusters using Penalty Parameter (λ)

Unlike K-Means, which requires pre-determining the number of clusters (K) using the Elbow Method, DP-Means is an effective option that optimizes the Penalty Parameter (λ).

- Logic of λ: This acts as a threshold that decides whether a new cluster will be created. If a customer's behavior is extremely distinct from the current cluster, meaning that their distance from existing centroids exceeds λ, the algorithm will detect and automatically create a new cluster.

- Experimental output: After tuning λ as opposed to forcing the data into pre-set groups, distinct customer groups were successfully discovered. This provides a significantly higher level of granularity, capturing subtle but important behavioral shifts, such as customers who spend a lot but only when they have a coupon, which can be hidden in larger categories.

The basic algorithm is presented in the code below:

```
def dp_means(X, lambd, max_iter=50):
    n_samples = X.shape[0]
    centers = [np.mean(X, axis=0)]
    assignments = np.zeros(n_samples, dtype=int)
```

The significant parameter is lambda (λ):

- In case distance > λ, start a new cluster.
- In case distance ≤ λ, then attach to the current cluster.

The data are rescaled prior to clustering:

```
features = ['Recency', 'Frequency', 'Monetary', 'Coupon_Ratio']
X_scaled = StandardScaler().fit_transform(rfm[features])
```

This guarantees that there is an equal contribution of all variables.

4.4.3. Experimental output: A general view of customer segments

Based on the clustering output, the dataset was automatically labeled into 8 distinct strategic groups:

Cluster ID	Behavioral Profile	Description
Cluster 0	High Recency / Low Frequency	**"Hibernating":** Customers that did not return in more than 250 days and showed low level of interaction.
Cluster 1	High Coupon_Ratio (> 0.65)	**"Deel seekers:"** Group that was sensitive with price. They only bought when there were coupons or discount promotions.

Cluster 2	Mid Recency / Mid Monetary	**“Stable”:** Average customers with predictive shopping habits, maintaining moderate spending and frequency levels.
Cluster 3	Extreme Monetary / Low Recency	**“Elite VIPs”:** They were the top 1% of spenders. They contributed high values without relying on coupons.
Cluster 4	Extreme Frequency / Low Recency	**“Power users”:** Highly habituated customers with extreme shopping frequency (weekly).
Cluster 5	Low Recency / Low Frequency	**“Newbies”:** Customers that have made recent purcharses.
Cluster 6	High Frequency / Mid Monetary	**“Loyalists”:** Consistent customers that purchased regularly every month. They helped to generate stable revenue fo the business.
Cluster 7	High Monetary / High Coupon_Ratio	**“Value-conscious high spenders”:** Group that was ready to pay a lot but was also aware of deals. They often wait for large promotions to place high-value orders.

Table 4.2: Behavioral Profiles of the 8 Statistical Clusters

4.5. Predictive models (LightGBM and CatBoost)

During this stage, the research witnesses a transition from static analysis to subjective prediction models. The main aim of this phase is to construct a proactive churn prediction system that can demonstrate the churn probability of customers based on their purchasing patterns. We have established a Comparative Study between two state-of-the-art algorithms: LightGBM and CatBoost. Instead of operating only one predictive model, performing two models at the same time can help us conduct a controlled experiment to evaluate how different gradient boosting architectures handle high-dimensional behavioral data and categorical variables.

4.5.1. Experimental environment setup

To ensure the competition between the two models was completely fair, the implementation followed a standardized multi-layer architecture:

- Stratified experimental splitting: The dataset was partitioned into 80% Training and 20% Testing. Stratified Sampling was used in this research to keep the churn rate (minority class) exactly the same in both datasets. Routinely having both datasets balanced keeps model bias from being introduced by a non-representative distribution of the target variable.

- Balancing the scales (SMOTE): Through the use of SMOTE (Synthetic Minority Over sampling Technique), the researchers were able to deal with an extreme class imbalance (the number of customers that "Stay" is substantially larger than those who "Churn"). Instead of simply duplicating pre–existing data using their technique, researchers used SMOTE to create "synthetic customers" based on what customers that have left look like in order to teach the model the abstract boundaries of "at-risk" behavior instead of simply memorizing the customers' history.

- Feature integration: The input for both models was a calculated mix of:

- Core RFM metrics: Recency, Frequency, and Monetary values.
- Behavioral reality: Coupon usage ratios, tenure months, and complaint history.
- Latent structures: Cluster labels from the DP-means algorithm, which help the models recognize complex, non-linear group structures.

4.5.2. Parallel model execution workflow

We employed two separate training pipelines to take advantage of each tree algorithm's unique strengths:
- The LightGBM pipeline was created using a leaf-wise expansion method for improved training speed and variance reduction. The tree is constructed by adding leaves to any branch that was last used to split on the maximum loss, meaning you can "dig deeper" into each branch and, therefore, discover the most complex patterns. Categorical variables were also manually encoded into numerical data before training with this model.

- The CatBoost pipeline was created using symmetric trees (i.e., all branches grow equally) with a focus on providing stability as well as directly handling categorical features. Both Cluster_ID and coupon status can be processed natively within the framework of CatBoost, without using traditional encoding methods, preserving the original dataset's structure.

Optuna was utilized for the processes of Bayesian Optimization, rather than manual tuning or exhaustive grid searching had been performed. As a result of this, Optuna automatically found the "golden" parameters of both models to ensure that their best performance was achieved prior to the final comparison:

- LightGBM tuning was centered around num_leaves, feature_fraction, and min_data_in_leaf.
- CatBoost tuning was centered around depth , l2_leaf_reg ,and random_strength.

4.5.3. Evaluation methodology

The more effective model is identified by not only simple accuracy, but also by three rigorous dimensions:

- Precision-recall curve (PRC): In churn prediction, the data is naturally imbalanced since there are far more customers who continue purchasing than those who leave. PRC helps to measure the model’s ability to catch churners while maintaining high confidence in its prediction. A high-performing model must catch as many at-risk customers as possible (high recall) without causing false alarms (high precision). This allows the business not to waste its marketing budget on discounts for customers who were never planning to leave in the first place.

- Computational complexity analysis: Even if a model is highly accurate can be useless in an e-commerce environment if it takes 10 minutes to produce a single prediction. Our group measured how long it takes the model to learn the entire dataset and the speed at which the model returns a result when a new customer transaction occurs. This is critical for real-world business use, where a company might want to send a retention offer the instant a customer shows signs of leaving.

- Evaluating XAI (Explainable AI) with a benchmarking technique: We utilized an interpretability method called SHAP (SHapley Additive exPlanations) to evaluate how two separate model architectures derive meanings from the same raw data. This is done to establish whether both algorithms recognize the same fundamental risk drivers (e.g. Complain or Tenure) and whether there is a divergence in the interpretation of the customer journey due to differences in their respective architectures. Independent identification of the same factors as high-risk by both models validates the integrity of our dataset.

1. Results and discussion

5.1. Descriptive statistics and RFM analysis

5.1.1. Demographic and Geographic overview

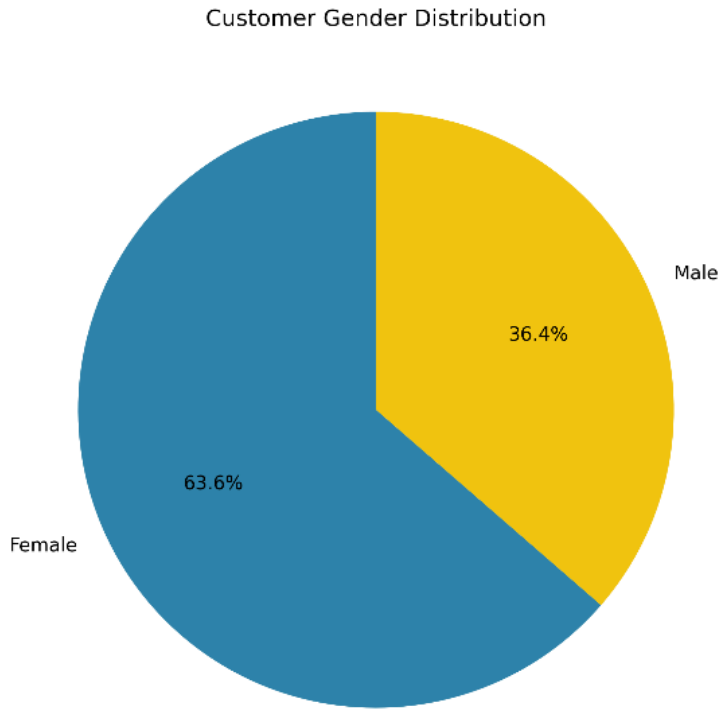


Figure 5.1: Customer gender distribution

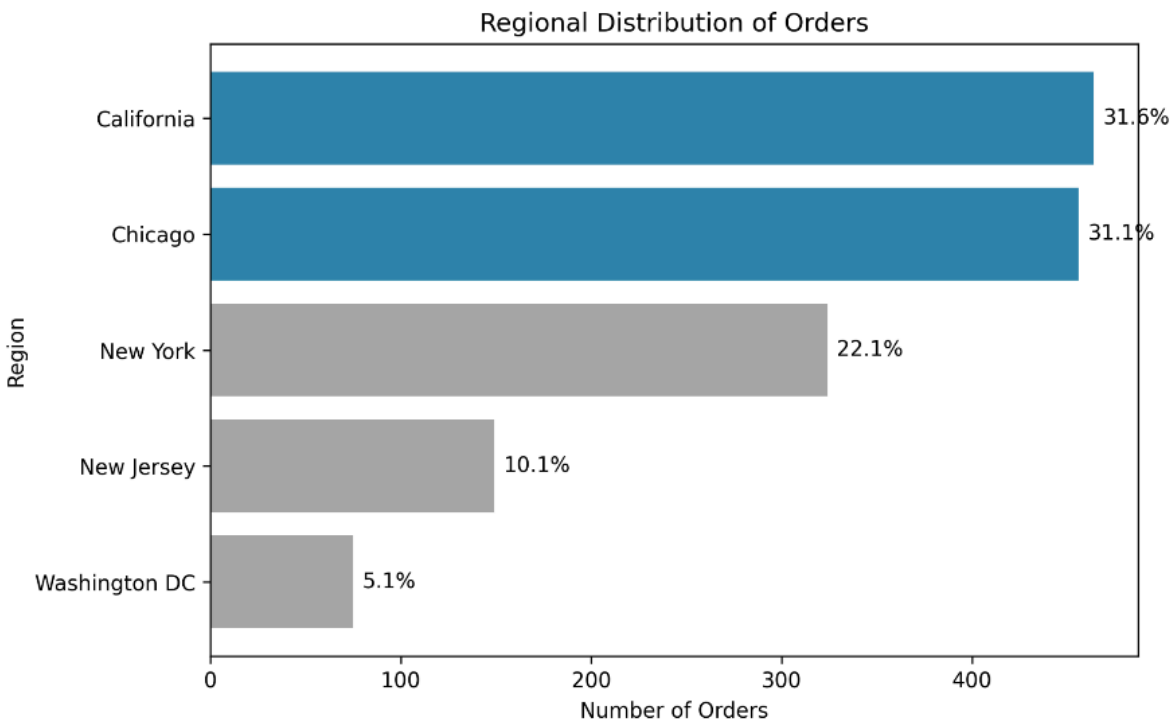


Figure 5.2: Regional distribution of orders

The data shows a slight tilt toward female customers, but it is not large enough to matter. Female customers make more purchases, but male customers generate more revenue, showing that fewer but higher-value transactions contribute more to total revenue.

Orders are concentrated in a few regions, especially California and Chicago. If activity drops in these areas, overall performance would likely be affected. In contrast, regions such as New Jersey (10.1%) and Washington DC (5.1%) have fewer orders per customer, which suggests weaker activity compared to the main regions. In contrast, New York (~22.1%) represents a middle region with moderate activity, and its customer behavior is not yet clear, so the region is an important feature for machine learning in the future.

5.1.2. Purchase behavior across product categories

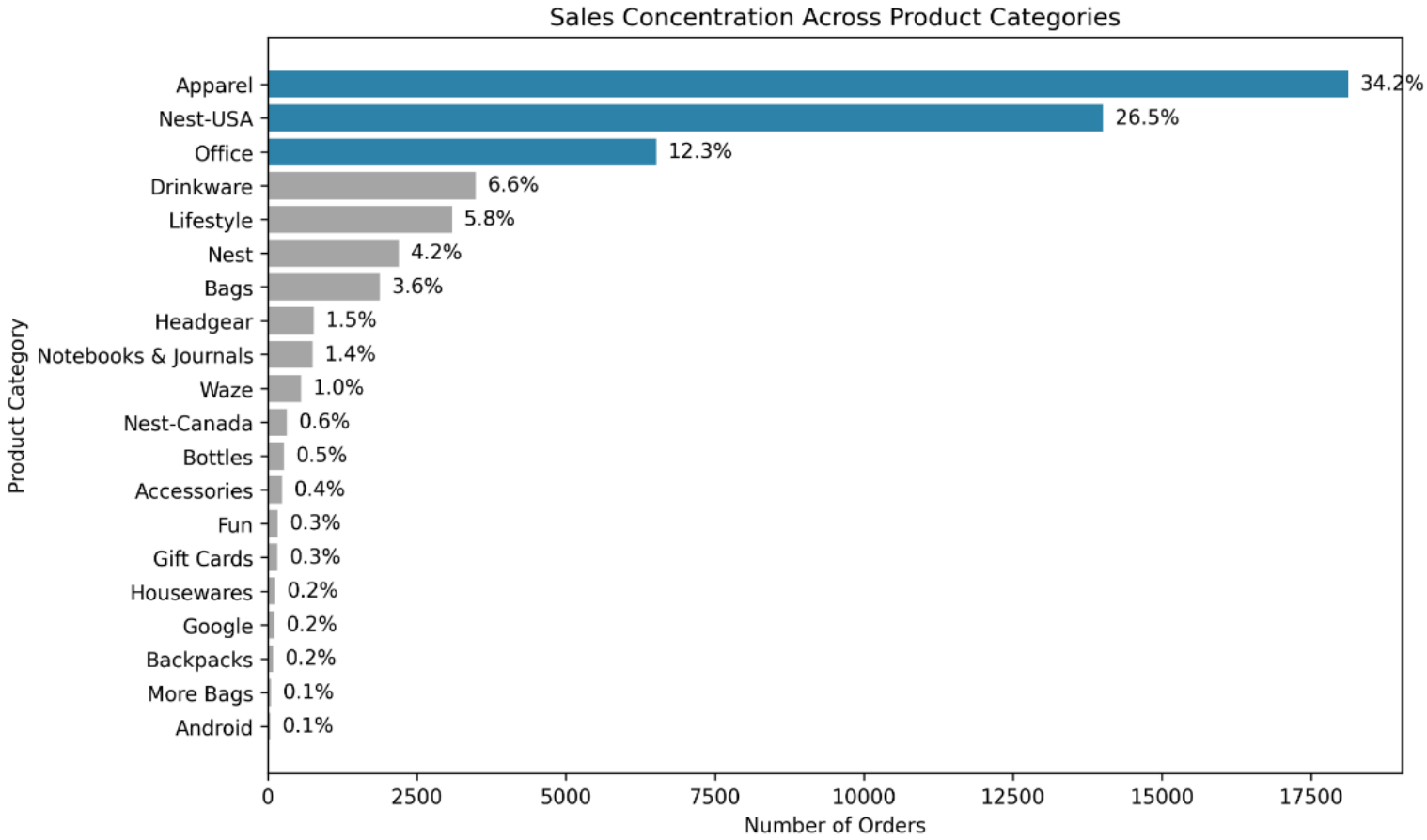


Figure 5.3: Sales concentration across product categories

Most orders come from a few product categories, so customer behavior differs across them. Customers who buy popular products and stay active are more stable, while those buying less common products with low activity leave more often.

5.1.3. Revenue structure and customer segmentation (RFM)

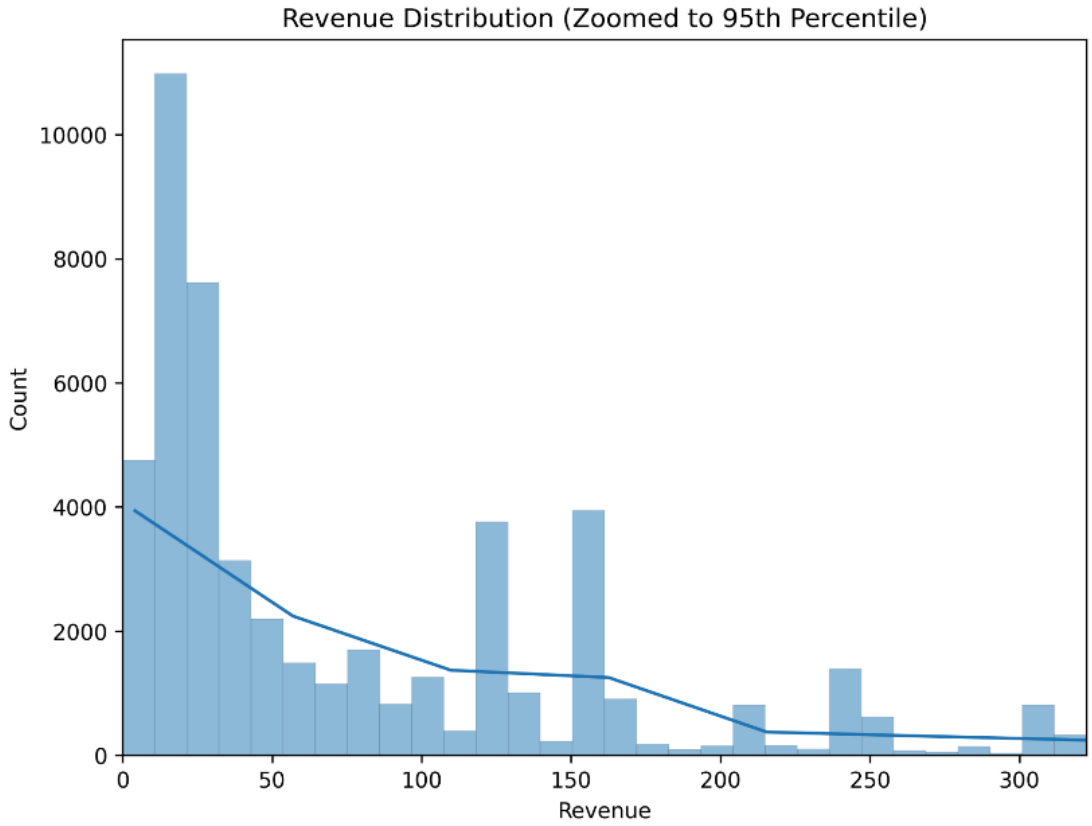


Figure 5.4: Revenue distribution (zoomed to 95th percentile)

The data shows a heavy-tailed distribution with a clear split between regular customers and VIPs. Most customers spend small amounts, this is the regular type. In contrast, a small group of customers spends much more and contributes a large share of the total revenue.

Churn from this high-value (VIP) group would have a much bigger impact on the business, while churn from low-value (regular) customers is less critical. Because of this, it is more useful to focus on value-weighted churn rather than treating all customers the same.

To support this, we add features Avg_Order_Value and Spend_last_30d in the data preparation for machine learning. For example, customers with high past spending but no recent activity are more likely to leave.

5.1.4 Churn pattern observations

Churn is taken from a separate dataset and is not connected to our main dataset, so it is used only for pattern observation.

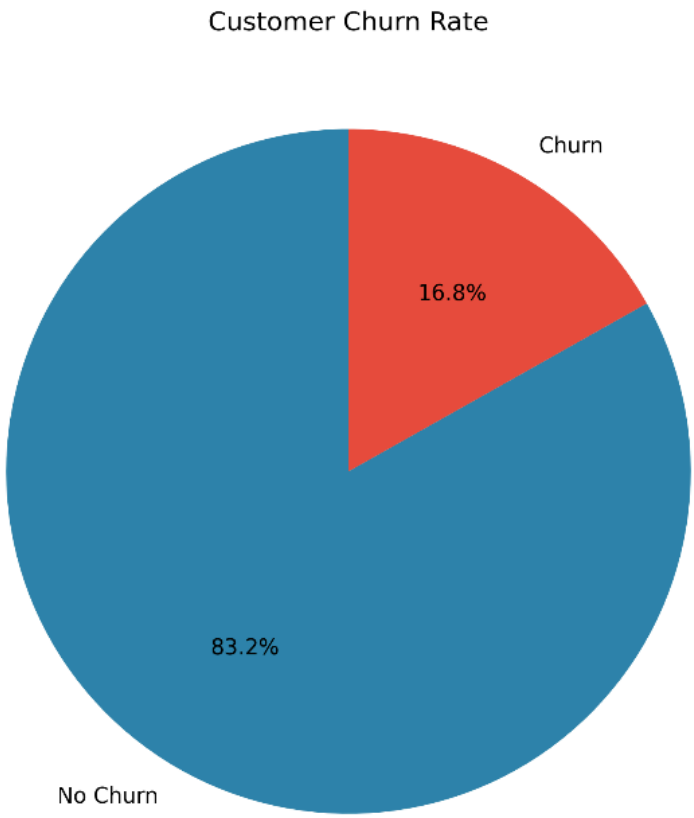


Figure 5.5: Customer churn rate

Only a small portion of customers leave, while most remain active. The limited number of churn cases makes differences less pronounced and harder to capture.

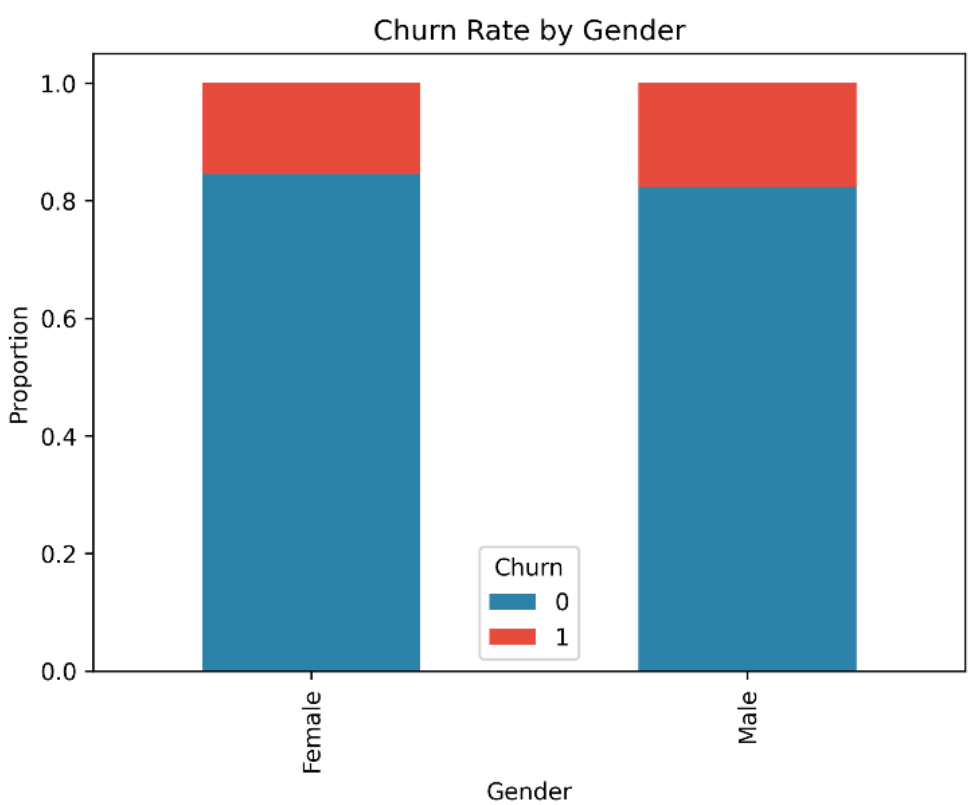


Figure 5.6: Churn rate by gender

Looking at churn by gender, there is no noticeable difference between male and female customers. The proportions are quite similar, which implies that gender alone does not seem to explain churn behavior in this dataset.

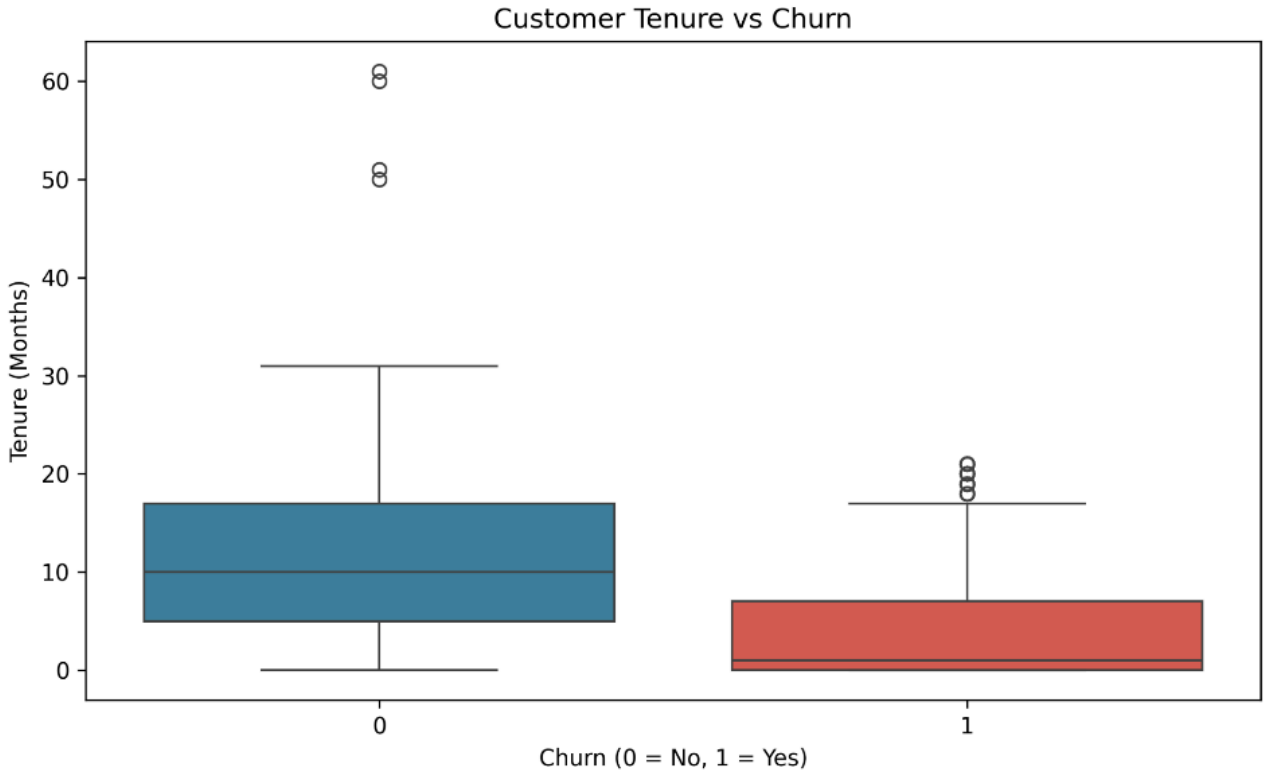


Figure 5.7: Customer tenure vs churn

However, the relationship between tenure and churn is much clearer. Specifically, customers who leave usually decide to churn in early stages, often within the first few months. In contrast, customers who stay tend to build up longer tenure over time. This gap is quite clear in the data and points to a weak early-stage retention, where customers drop off before becoming regular users.

These patterns are useful for understanding customer behavior, although they are explored separately since this dataset is not directly connected to the main transaction data and cannot be reliably combined.

5.1.5. Real business implications

The exploratory analysis suggests several patterns that are relevant for business decision-making.

Most churn happens early. The problem is in the early stages. Onboarding and early engagement matter the most here.

A small group of customers brings in a large share of revenue. Losing even a few of them would hit the business more than losing many regular customers. Retention should focus more on these high-value customers.

Customer activity is concentrated in a few regions, especially California and Chicago. A few regions drive most of the revenue. If these areas slow down, overall revenue will drop. At the same time, lower-activity regions show weaker engagement and may require different strategies.

In conclusion, behavioral factors such as tenure and recent activity appear to be more important than demographic factors like gender. This suggests that business decisions should rely more on customer behavior rather than basic customer profiles. Overall, the main issue is not just churn itself, but where and when it happens, as well as which customers are most valuable to retain.

5.2. Inferential statistics

5.2.1. Independent T-test: The impact of coupons on Average Order Value

Considering T-test findings, we can see that the p-value is 0.00126. This value is less than the significance level of 0.05 and this implies that the null hypothesis is not accepted. Based on the visualization (bar chart of AOV) it is evident that there is a higher value in order amongst customers who use the coupons as opposed to those who do not.

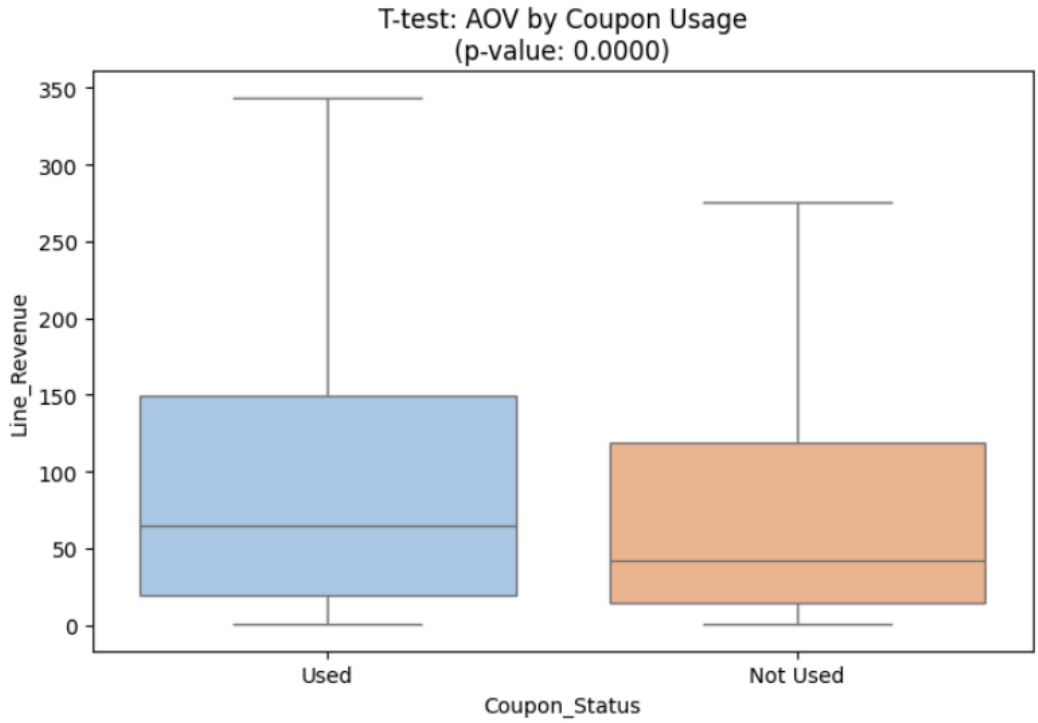


Figure 5.8: AOV by Coupon Usage

5.2.2. ANOVA: Efficiency analysis across discount tiers

In terms of the ANOVA test, the p-value 0.007 is statistically significant ($p<0.05$), which means that there are indeed significant differences in revenue across the different discount levels. This is confirmed by the view, which demonstrates different discount levels (10%, 20%, and 30%) have varying impacts on revenue, rather than a consistent impact.

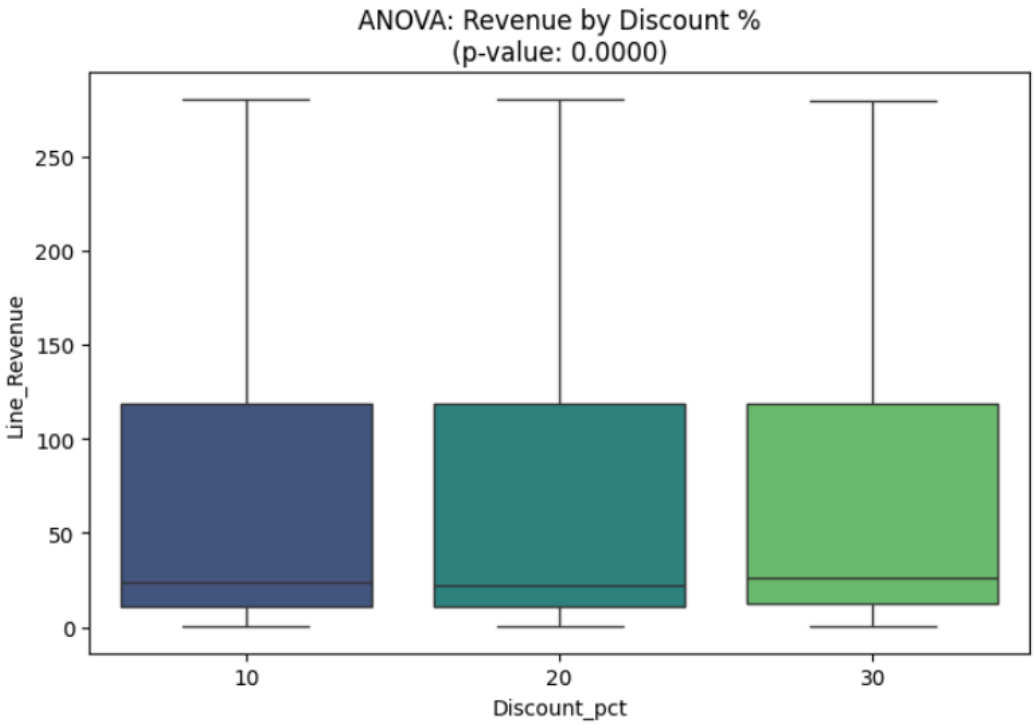


Figure 5.9: Revenue by Discount %

5.2.3. Chi-square test: Linking service friction to retention

In the Chi-square test, the p-value is very insignificant (2.66 1078). The stacked bar chart shows that the churn rate of customers who complain is by far on the higher side (31.7) than with the customers who do not complain (10.9).

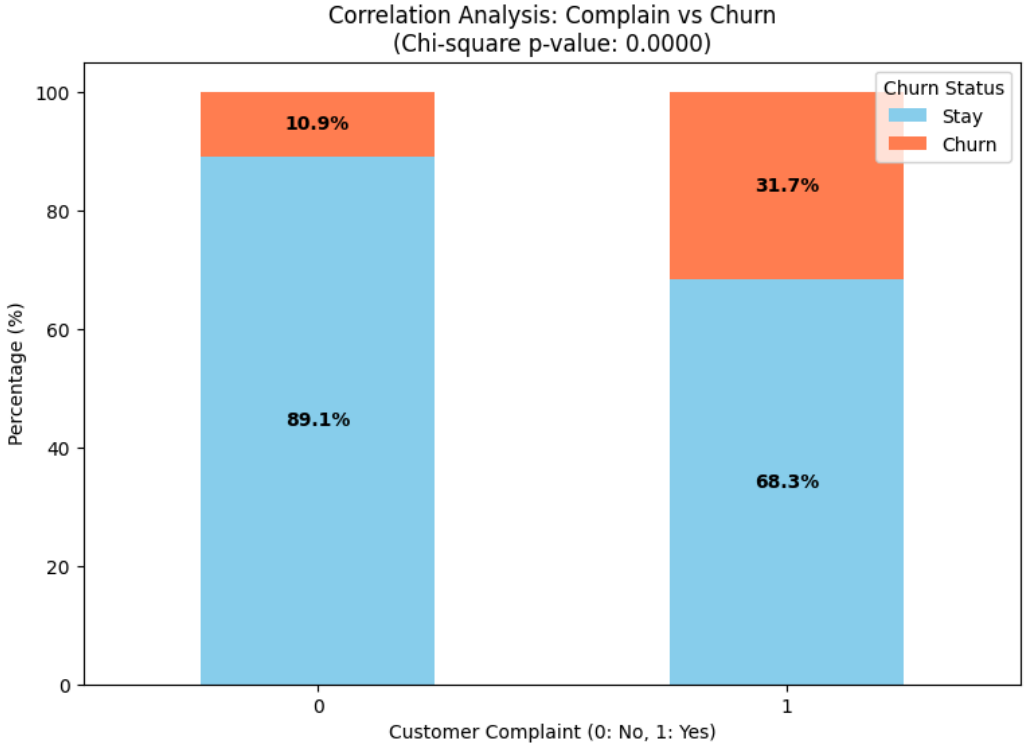


Figure 5.10: Correlation analysis: Complain vs Churn

In this visualization, it is clear that the churn rate of customers who complain is significantly higher.

5.2.4. Discussion and insights

Based on these findings, it is apparent that promotions are highly influential in customer behavioral change in terms of expenditure. Coupons have the opportunity to boost AOV, a good move when revenue needs to be boosted in the short-term. A result of the ANOVA would indicate, however, that all levels of discounting are not equally successful and therefore a business must select the most appropriate level.

More to the point, Chi-square outcome reveals that experience in the eyes of customers is a critical factor. Although promotions may boost expenditure, they will not avert churn in case customers are not satisfied. The complainations are good indications of dissatisfaction among customers and thus should be addressed promptly.

Real business implications:

- Marketing & sales strategy: The statistics demonstrate that coupons are good at extending the wallet of the customer in the short run, resulting in an increase in AOV. But the ANOVA results indicate that the 30 percent discount level may not increase the revenue proportionally in relation to the 20 percent discount, thus suggesting that the more discounts are given the less incremental revenue will be obtained.

- Customer retention: The most important business lesson is that poor service cannot be covered with coupons. A negative experience (complaint) is the main reason for churn even when a customer uses a coupon. Thus, the "Loyalty" gained due to discounts is not so strong and secondary to Loyalty gained due to the quality of the service.

- On balance, the inferences made by inferential statistics are quite convincing that the promotion strategy and service quality matter, although service quality has a greater influence on long-term loyalty.

5.3. Lasso Regression

5.3.1. Output and results

A score of 0.98 in the Lasso model is a result representing excellent predictive performance of the model.

As indicated by the coefficients of the model, Frequency is the only model coefficient that is positive and significant (+0.2227) whereas Recency, Monetary and Coupon Ratio were brought to zero. This implies that in this model, the variables are irrelevant in predicting customer loyalty.

The significance of each feature is vividly demonstrated in the output table, and multiple visualization of the coefficients demonstrates that Frequency is predominant.

The high predictive accuracy of this model is proved by its AUC-ROC of 0.98. The most interesting result was the reduction of the Monetary and Coupon Ratio to zero, which left Frequency (+0.2227) being the only dominating predictor.

5.3.2. Discussion and insights

The greatest point in this model is that customer loyalty highly depends on purchase frequency. This implies that those customers who shop frequently will have a higher chance of remaining loyal, no matter the amount that they use within a given visit. This is contrary to the popular belief whereby the common attitude of businesses is centering on the high spenders. Nevertheless, Lasso model indicates that repeat behavior is a key factor as compared to high value purchases only once.

The other benefit of this model is simplicity. Lasso eliminates irrelevant variables to generate a clear and readable model. This facilitates easier interpretation of the results by managers and their application in the actual business strategies.

This finding also complements the segmentation analysis, in which loyal customers will be characterized by unchanging purchasing behavior.

Real business implications:

- Frequency vs Monetary debate: This is a paradigm shift in sales terms. The habit of purchase (Frequency) defines true loyalty in this project and not the size of purchase (Monetary).

- Coupon strategy optimization: The shrinkage of the Coupon_Ratio to zero means that coupons do not provide good predictability of high valuation in the long-term. Actually, the high usage of coupons could be typical of deal-seekers, but not loyalists.

- Strategic recommendation: The marketing teams ought to change their KPIs to be Total Revenue per Sale to Customer Purchase Interval. The number of visits should be rewarded in the form of the loyalty programs and not the amount spent.

5.4. Segmenting customers using DP-Means Clustering

5.4.1. Results and visualization

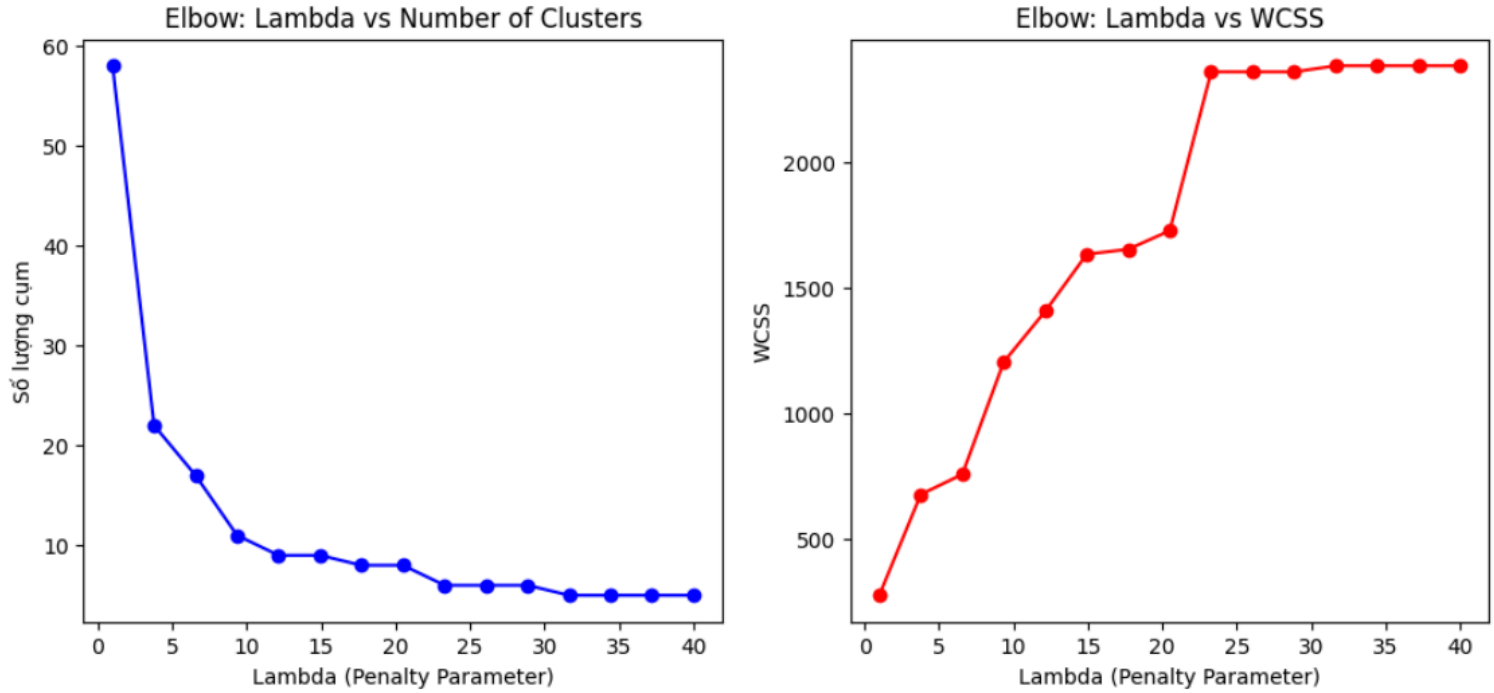


Figure 5.11: Elbow chart

The Elbow chart (Lambda vs Number of Clusters and WCSS) was created by trying various values of lambda hence Figure 5 was created. The finding indicates that optimum λ is around 15-20.

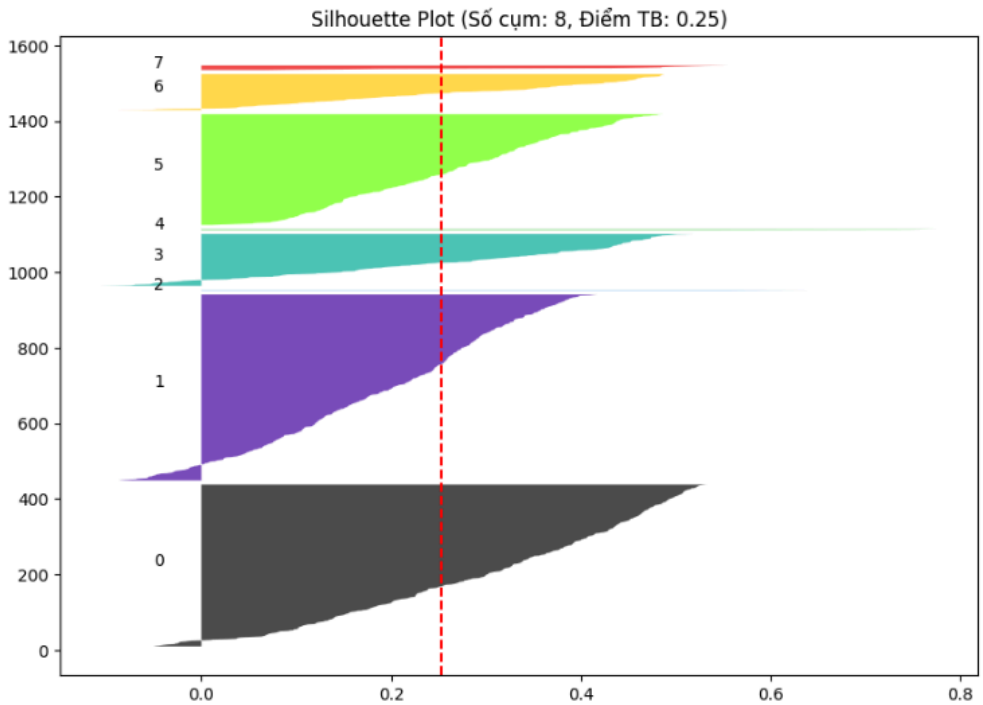


Figure 5.12: Silhouette plot

The score (-0.25) shows high quality of clustering in medium.

The DP-Means algorithm found 8 clusters that represent various customer-behavior types. These include:

- High frequent and high spending VIP customers.
- Recurring and steady buyers.
- Buyers of high coupon utilization.
- Potential or New customers.
- Low-activity at-risk customers.

The Elbow Plot suggests that a λ value of 15 is the most suitable for the model to achieve balance and accuracy. The Silhouette value of 0.25 indicates a degree of separation between the clusters. In addition, looking at the visual representation of the clusters shows that the two groups differ greatly from each other with respect to how often each group used the coupons provided. The resulting clusters provide an acceptable amount of separation based on the model (Silhouette Score 0.25). The two groups were segmented as follows: groupings of Clusters 3 and 4 (organic VIP cluster) versus Cluster 1 (deal hunters). Cluster 1 represents individuals with a high coupon ratio, and therefore they would only generate volume if they were offered a discount. Clusters 3 and 4 have a low coupon ratio and are capable of generating significant revenues organically.

5.4.2. Discussion and insights

The primary advantage of DP-Means is its flexibility. It is able to learn natural tendencies in the data without strongly arming it into a definite number of clusters. This yields more realistic and significant segmentation.

A lesson that can be learned is that not every high-value customer is the same. There are those customers who end up paying much but without the use of the discount among others and there are those who are over reliant on promotions. The similar treatment of such groups may result in ineffective strategies.

Customer behavior is yet another key finding which is multidimensional. Frequency and coupon usage, as other factors offer better understanding than the value of money.

In terms of business, this type of segmentation enables businesses to draw specific strategies. As an example, VIP customers might want other exclusive services whereas deal seekers are more likely to be attracted to discounts. High-risk-customers should be reactivated through reactivation campaigns.

Real business implications:

- Margin optimization: The discovery of Value-Conscious High Spenders (Cluster 1) is a key practical implication. There is no use in spending marketing budgets to continue giving deep discounts to Organic VIPs (Clusters 3 and 4) since they would buy anyway.

- Targeted retention (Win-back): In the case of the At-Risk segment (Cluster 0), the model would have the coupons as a key reactivation tool. And here the "Impact of Promotion" is greatest, not in gratifying the already true believers, but in converting the hibernated.

- Personalization of sales: Sales teams can apply such cluster labels to customize communications. To illustrate, the emails of "Deal Hunters" should be overloaded with promotion, whereas a message of "Champions" should be based on status and not on price and should be titled as an Early Access or Exclusive Service.

5.5. Predictive models (LightGBM and CatBoost)

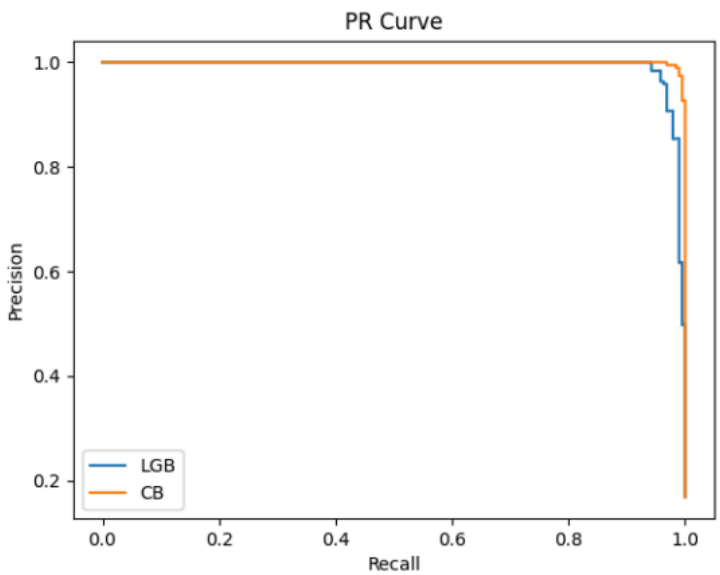
5.5.1. Performance evaluation

LightGBM shows very high performance on the test set, with results close to the best possible level.

- ROC-AUC is 0.9976, which shows that the model can separate churn and non-churn customers very well.
- PR-AUC is 0.9922, which shows that the model handles imbalanced data effectively. In churn prediction, PR-AUC is more important than ROC-AUC because it focuses more on churn customers.
- Precision is 0.9838, Recall is 0.9579, and F1-score is 0.9707. This means the model is both accurate and well balanced in detecting churn while reducing wrong predictions.

CatBoost shows slightly better results than LightGBM.

- ROC-AUC reaches 0.9999
- PR-AUC reaches 0.9993
- Precision, Recall, and F1-score all reach 0.9895



PR Curve of CatBoost is almost above LightGBM across the entire recall range, which shows better and more stable churn detection performance.

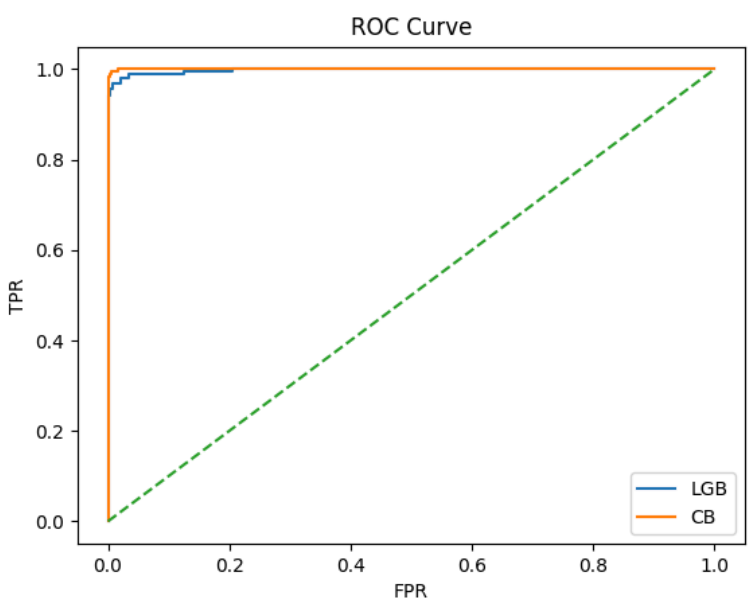


Figure 5.13: ROC curve and PR curve

5.5.2. Threshold optimization

LightGBM does not use the default threshold of 0.5. Instead, it uses the optimal threshold:

Threshold = 0.4141

Reducing the threshold helps to:

- increase Recall
- reduce false negatives
- avoid missing churn customers

This result matches the PR Curve shape, where precision is still kept at a high level across most recall values.

In real business situations, missing customers who may leave causes more loss than predicting some customers incorrectly. Therefore, this threshold is more suitable for customer retention strategies.

The optimal threshold of CatBoost is:

Threshold = 0.1616

This threshold is much lower than LightGBM, which helps to:

- increase churn detection
- maintain high precision
- better support retention strategies

This shows that CatBoost has very strong prediction quality even with a low threshold

5.5.3. Model behavior analysis (SHAP)

- *LightGBM*

The most important features are:

- Tenure
- Complain
- NumberOfAddress
- CashbackAmount
- SatisfactionScore

Among them, Tenure is the most important factor.

- Low Tenure → higher churn
- Having complaints → higher churn

In addition, SatisfactionScore and DaySinceLastOrder also affect churn. Higher SatisfactionScore is linked to higher churn, although this is unusual. This may mean that short-term satisfaction does not always lead to long-term loyalty. Lower DaySinceLastOrder is also linked to higher churn. This shows that some customers may leave even after buying recently. Their buying behavior may be temporary, not stable.

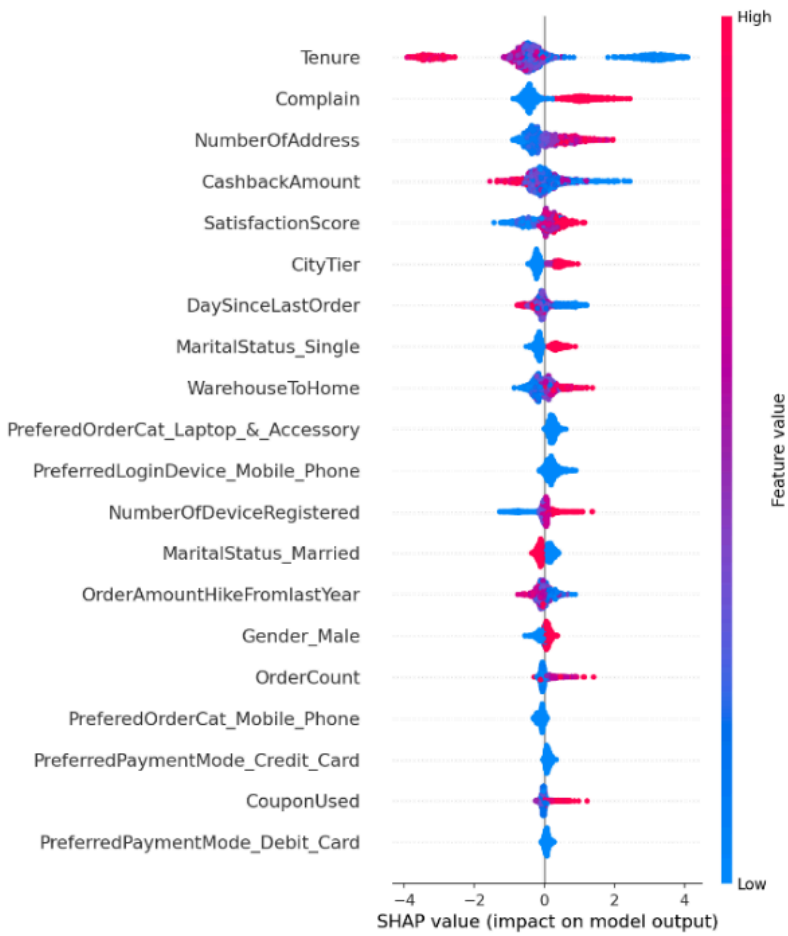


Figure 5.14: Importance of features in LightGBM

- *CatBoost*

The top features of CatBoost are:

- Tenure
- Complain
- NumberOfAddress

- SatisfactionScore
- CashbackAmount

The results show that:

- Low Tenure → higher churn
- Having complaints → higher churn
- Higher SatisfactionScore → higher churn
- Lower DaySinceLastOrder → higher churn

CatBoost shows a more stable SHAP distribution, especially for categorical features because it can handle categorical features directly without the need for encoding.

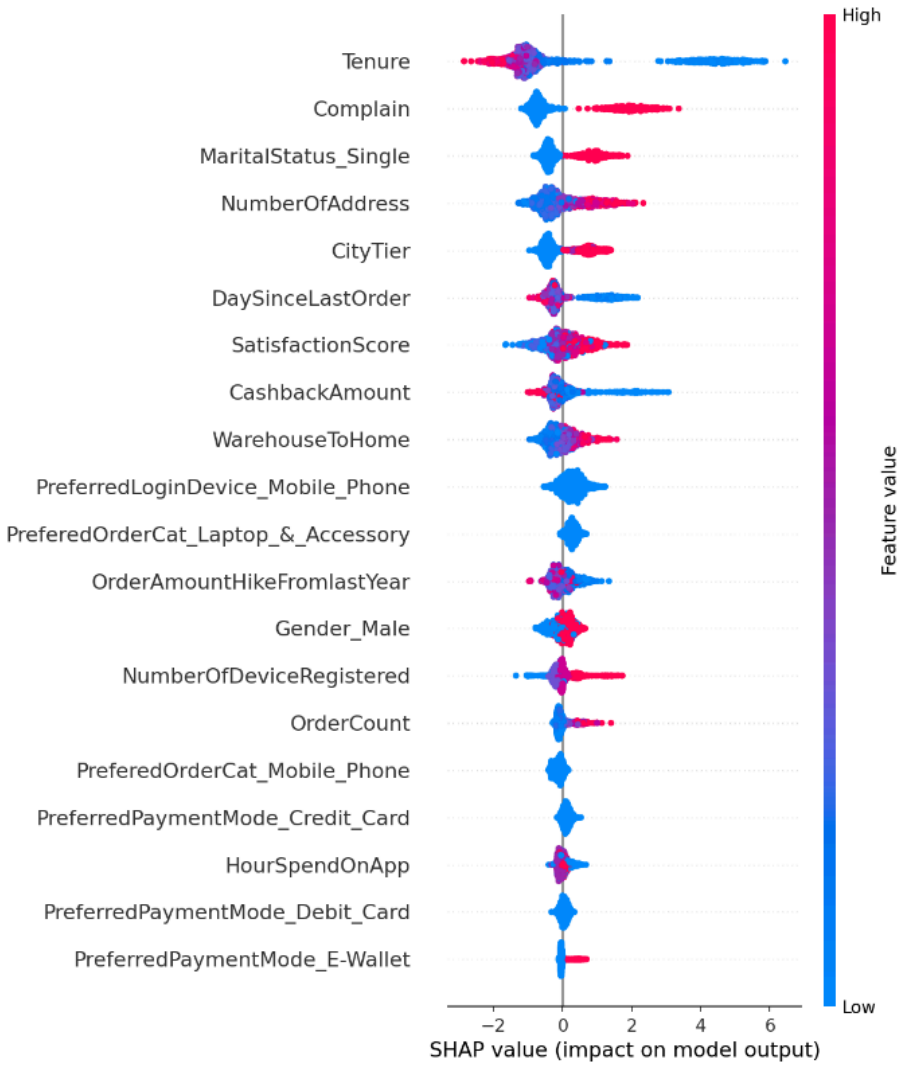


Figure 5.15: Importance of features in CatBoost

5.5.4. Discussion

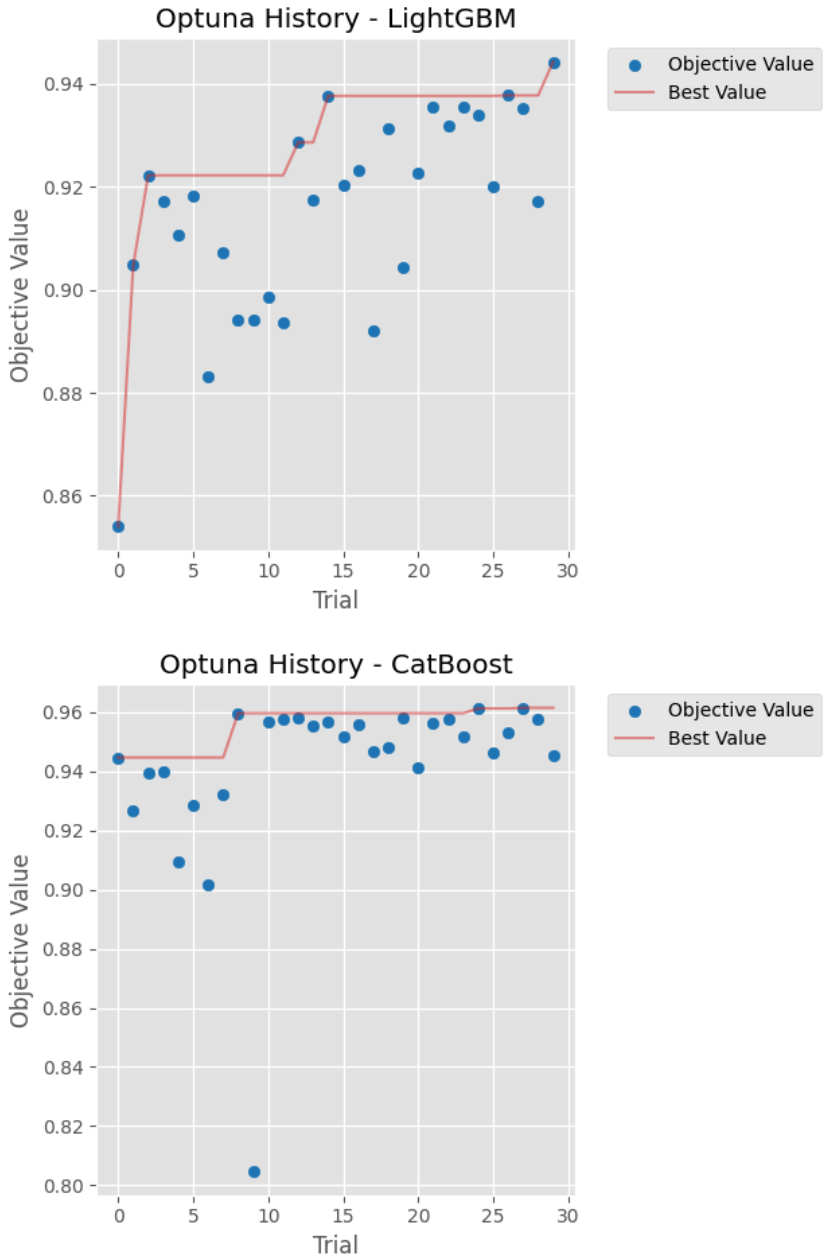


Figure 5.16: Optuna history of LightGBM and CatBoost

LightGBM has several advantages (see Figure 3 in Section 4.4):

- fast training speed
- easy hyperparameter tuning
- good performance with large datasets

However, the very high performance also raises some concerns about:

- overfitting
- data leakage
- data that may have very clear class separation

One-Hot Encoding has been substituted for Label Encodings of categorical variables in the final model and allows for greater separation of categories so that they do not mix together during the learning process in the final model. However, since One-Hot Encoding adds additional dimensions to the input space compared with label encoded variables, this could potentially increase the complexity of the model. In spite of the complexities found within the model, the SHAP results indicate the model can still learn to recognise normal business patterns; thus demonstrating high levels of predictive capability in practice.

CatBoost has several advantages:

- better handling of categorical features
- no need for manual encoding
- better generalization ability
- more stable performance

Compared with LightGBM, CatBoost depends less on preprocessing and reduces the risk of errors from feature engineering. Although the very high performance still needs overfitting checks, the SHAP results show that the model learns real churn behavior.

5.5.5. Model comparison

Both models work quite well, although CatBoost works a little better in:

- PR-AUC
- Remember
- Score F1

LightGBM is better at training quickly and adjusting hyperparameters in a flexible way. CatBoost is better at dealing with categorical features and keeping models stable. Although both models achieve strong performance, CatBoost demonstrates better handling of categorical features, while LightGBM shows slightly faster convergence. Overall, CatBoost is the more suitable choice for churn prediction in this study.

5.5.6. Real business implications

Focus on new customer experience

- SHAP shows that Tenure and Complain are the most important factors affecting churn.
- New customers usually have low loyalty and are more sensitive to service problems.
- If a customer makes a complaint early, there is a high chance they may leave the service.

Grasp short-term satisfaction

- Some groups of consumers will have higher turnover based on their Happiness Score in relation to how long it has been since they have made a purchase (i.e., short amount of time since last transaction).
- This suggests that individuals can be satisfied at the point of purchase but may not be loyal after they leave.
- Being temporarily satisfied does not guarantee long-term loyalty

Value of financial benefits

- The Cash Back Amount is one of the most critical components in the CatBoost model.
- Financial benefits like cash back and discounts have a significant impact on the decisions of some customer segments.
- Financial benefit will ultimately guide their decision to either remain or leave.

Proactive retention strategies

- The low optimal thresholds (0.1616) reflect that Recall is weighted much higher than Decay in the model.
- Faint signals of churn can help highlight the danger of potential churn at an early stage.
- Preventative measures taken to prevent customers from churning will typically cost less than acquiring new customers.

1. Conclusion and recommendations

6.1. Conclusion

Through a careful process of testing and looking at many sets of e-commerce data, we have reached some crucial conclusions about modern customer behaviors and the relationship between coupons and loyalty. Most importantly, we have learned that customer loyalty will not be defined solely by static attributes (gender, location, etc.) but rather through the behaviors that customers demonstrate, such as the length of time they have purchased from a particular brand or the manner in which they access discount coupons.

We were also able to determine from our data that there exists a "high-risk" time frame for customers during the first few months of their relationship with the retailer. This is the time when customer engagement is very fragile and where attrition rates are at the highest point; however, once the customer has successfully surpassed this critical point, the likelihood of them becoming a long-term regular customer is multiplied many times.

Additionally, when we looked at the revenue distribution from a financial structural perspective, identifying the "heavy-tailed" revenue distribution has significantly changed our approach to risk management. The loss of one VIP customer through our use of DP-means clustering causes much more damage to cash flow than would the loss of one "regular" user.

Finally, while LightGBM has exhibited superior processing speed and computational efficiency, CatBoost was found to be much more sensitive in processing complex categorical variables. This allowed it to capture high-risk churn scenarios with greater precision without sacrificing the original integrity of the data features.

6.2. Recommendations

Our strategic roadmap aims to deliver three pillar-based roadmaps for maximising revenue sustainability and brand health by providing the best service to our core customers through 1) transitioning to value-based retention instead of mass promotions, 2) implementing a first 90 days engagement programme, and 3) implementing CatBoost and LightGBM dual-engine technology for retrospective and real-time retention interventions.

In order to improve brand health and long-term revenue sustainability, it is vital to stop mass promotions and start focusing on value-based retention by investing marketing dollars into the VIP groups and using non-monetary perks such as preferential support/early access instead of monetary rewards as your means of differentiation.

To resolve early attrition, the organisation should establish a 90-day engagement plan that uses the first 90 days to establish customer loyalty via improved onboarding and behavioural nudges.
Thirdly, from an operational technical perspective, we recommend CatBoost for high-accuracy strategic planning at set intervals and LightGBM for real-time strategic planning to deliver instant, effective retention offers during live user interactions.

We do not recommend pursuing aggressive, large-scale investments in new (undefined) markets such as London and New York City. Instead, it is wiser to begin with small-scale A/B testing to uncover the specific "growth formula" for that region. By tightly integrating artificial intelligence with decisions based on actual behavioral data, the business can ensure it not only retains its most valuable customers but also optimizes every dollar of its marketing expenditure.

6.3. Limitations

The most obvious limitation is the narrowness of the secondary dataset. This means that transactional data can show us things like transaction information and how often coupons are used, but it can't tell us why customers choose to buy anything. We can see that a customer in the At-Risk cluster has not purchased for more than 250 days, but we cannot determine whether it is because they are dissatisfied with the product and service, or whether it is simply because of a promotion provided by competitors. The research has not approached qualitative “Customer Sentiment” data, personal motivation, and the real level of satisfaction among customers. As a result, we are observing the symptoms of loyalty, rather than the root cause.

Furthermore, our models operate within a closed system. They do not consider external contributors to customer loyalty from the dynamic market, such as economic inflation, marketing strategies from competitors, or seasonal fashion shifts.In actuality, a positive coefficient for Frequency could be very different if there is a sudden problem in the supply chain or a bad review goes viral.

Another boundary of this study is that we are using the past to navigate the future.RFM analysis, the Lasso Regression model, and predictive machine learning models now rely on previous transactional data. But in the fast-paced world of e-commerce, people's shopping preferences change all the time. A static model that is not fed with a real-time data stream can only provide accurate answers to questions that are no longer relevant. This leads to a requirement that these models need to be retrained periodically to maintain their accuracy.

6.4. Future research

The most immediate opportunity lies in addressing the gap between transactional actions and motivations. One potential direction of investigation is the incorporation of other and finer measures like customer reviews and satisfaction surveys into the RFM-C model. Understanding the perception of customers regarding their spending will allow the examination of the paradox of satisfaction and churn: why satisfied customers, who use coupons and have positive perceptions about the products, end up churning. It may be found that even with a high satisfaction score, the customer continues to churn because of service friction, economic constraints, or a change in their lifestyle towards deals.

In addition, to address the inherent limitations of the existing models due to their static nature, future research must move toward models that adapt to changing environments dynamically. A move away from static regressions to recurrent neural network structures such as LSTM (Long Short-Term Memory) networks or GRU (Gated Recurrent Unit) networks would help in predicting customer churn in real time. Such an analysis would be useful for businesses as they could then take proactive measures like issuing a coupon when the likelihood of a customer engaging diminishes.

Beyond binary classification (Churn vs. Non-churn), future studies should focus on the financial lifespan of the customer. We can focus on predictive CLV modeling. For instance, using Gamma-Gamma or Beta-Geometric (BG/NBD) models to estimate the specific monetary value a customer will generate over the next 24 months. This would

allow for a precise calculation of the Targeted CAC (Customer Acquisition Cost), ensuring that coupon investments never exceed the predicted return from a specific segment. We also need to research the exact point where a discount is high enough to encourage a customer to buy, but not so high that the business loses profits. This allows for "scientific discounting", meaning that giving a 5% discount to one group and 15% to another based on what actually works for them.

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TASK DISTRIBUTION TABLE

Full name	Responsibilities	Completion
Lữ Lê Hương Giang	- Team leader - Handle these methods and models: Inferential statistics, Lasso regression, DP-Means	100%
Nguyễn Hà Linh	- Provide necessary information to write report - Handle predictive models: LightGBM and CatBoost - Provide necessary information to write report	100%
Trần Minh Vy	- Pre-process data by creating cleaned datasets - Calculate R, F, M variables and Coupon_Ratio - Handle the descriptive statistics and RFM analysis	100%
Nguyễn Lê Hoài An	- Provide necessary information to write a report - Gather provided information - Write the full report - Write the IEEE paper - Check AI generated content and plagiarism in the final stage.	100%